1. [generating functions in probability] I throw a fair coin until the sequence HHH appears.

   a) Draw the directed graph for this game.

   b) Setup a system of equations for the sequences with given endpoint and starting point at the start of the game.

   c) Explain how would you calculate the expected number of tails from the start till the end of the game. Note: you don't need to do actual calculations nor to solve the system, just describe how can you extract the expected value using derivatives.

2. [Polya-Redfield counting] Consider a 3×3 mosaic of gems, where each gem may be of one of k possible types. We consider mosaics identical if the can be transformed one into another by rotation. Flips are not available!

   a) List all elements of the group of actions that act on mosaics.

   b) Write each action as a combination of cycles and provide its cyclic monomial.

   c) Find the cyclic index and the number of different possible mosaics for $k = 10$ gem types available.

3. [planarity and Euler formula] Rhombicosidodecahedron has 20 triangular faces, 30 square faces and some number of pentagonal faces. The total number of vertices is 60.

   a) How many pentagonal faces does rhombicosidodecahedron have?

   b) How many edges are meeting at every vertex?

4. [graph colouring] Consider the graph with three components. The first component is a C_4 graph, the second one — a tree with 10 vertices, the third one — an isolated vertex. Components are non connected.

   a) Draw an example of such a graph.

   b) You have 10 colours available. In how many ways can you color the vertices of this graph if adjacent vertices should be painted differently?

5. [linearity of expected value] We put randomly 7 pencils in 10 boxes one by one choosing each box with equal probabilities.

   a) What is the average number of empty boxes?

   b) What is the average number of boxes with more than one pencil?

6. [old topic] The generating sequence $g(t)$ of an integer sequence $(c_0, c_1, c_2, \dots)$ satisfies the equation

$$g(t) = 1 + t \cdot g(t) \cdot g(t).$$

   (a) Solve the equation for $g(t)$.

   (b) Which of the two roots should we choose if all c_n are positive?

   (c) Find c_3.